Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology

B.Tech. | Sem-5 | Winter-2025

Course Code	:	2301CS511	**Date**	:	17-10-2025
Course Name	:	Advanced Web Technology	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain useMemo hook in ReactJS.	**4**
	(B)	Write a short-note on useFormStatus hook.	**3**
		OR	
		Write a short-note on useCallback hook.	
	(C)	Write a code to create a custom hook in ReactJS to perform fetch API operation.	**7**
		OR	
		Explain useDeferredValue hook in ReactJS with example.	
Q.2	**(A)**	Explain local state management in ReactJS.	**4**
	(B)	Write a short-note on managing global state using context API.	**3**
		OR	
		Write a short-note on state management approaches.	
	(C)	Write a program in ReactJS to manage student state in store using Redux.	**7**
		OR	
		Write a code to create basic chat application using socket.io	
Q.3	**(A)**	Explain how to manage status code in NestJS.	**4**
	(B)	Write a code in NestJS to perform CRUD operation on a table with RoomID, RoomName, RoomFloor, RoomCapacity, RoomFare, isAvailable fields.	**10**
		OR	
		Write a code in NestJS to get available rooms with fare between given range on a table with RoomID, RoomName, RoomFloor, RoomCapacity, RoomFare, isAvailable fields.	
Q.4	**(A)**	Explain Route Group in NextJS.	**4**
	(B)	Write a short-note on nested layout in NextJS.	**3**

OR

List any six file conventions in NextJS.

	(C)	Explain Dynamic Route in NextJS with example.	**7**

OR

Explain Rendering Fundamentals in NextJS.

Q.5	**(A)**	Explain basics of fetching data in NextJS.	**4**
	(B)	Write a code in NextJS to perform getAll, getByID and delete operation on a table with RoomID, RoomName, RoomFloor, RoomCapacity, RoomFare, isAvailable fields.	**10**

OR

Write a code in NextJS to perform getAll, and insert operation on a table with RoomID, RoomName, RoomFloor, RoomCapacity, RoomFare, isAvailable fields.

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